Electronics Coming To The Aid Of Failing Human Body Parts

With the advances in science and technology, there has been a phenomenal growth in the use of electronic devices in the medical field. At present, various electrical devices are being engineered that could help patients recover from illness by replacing almost any disabled body part

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Science and technology are making great advances in all fields, particularly with the phenomenal growth of applications in various fields. The medical field has also embraced the use of electronic devices by not only helping patients recover from illness but also replacing medically stained/disabled human body parts, thus moving towards a complete electronics human body.

At present, various electrical and electronic devices are being engineered that can blend with the organs and tissues of the human body.

Some of the commonly used electronic devices at present are mediators, recorders, and stimulators of electricity, with the capacity to monitor important electrophysiological events, replace disabled body parts, or even stimulate tissues to overcome their limitations. The use of such electronic devices or organs in place of disabled human organs is capable of leading humans into the age of cyborgs.

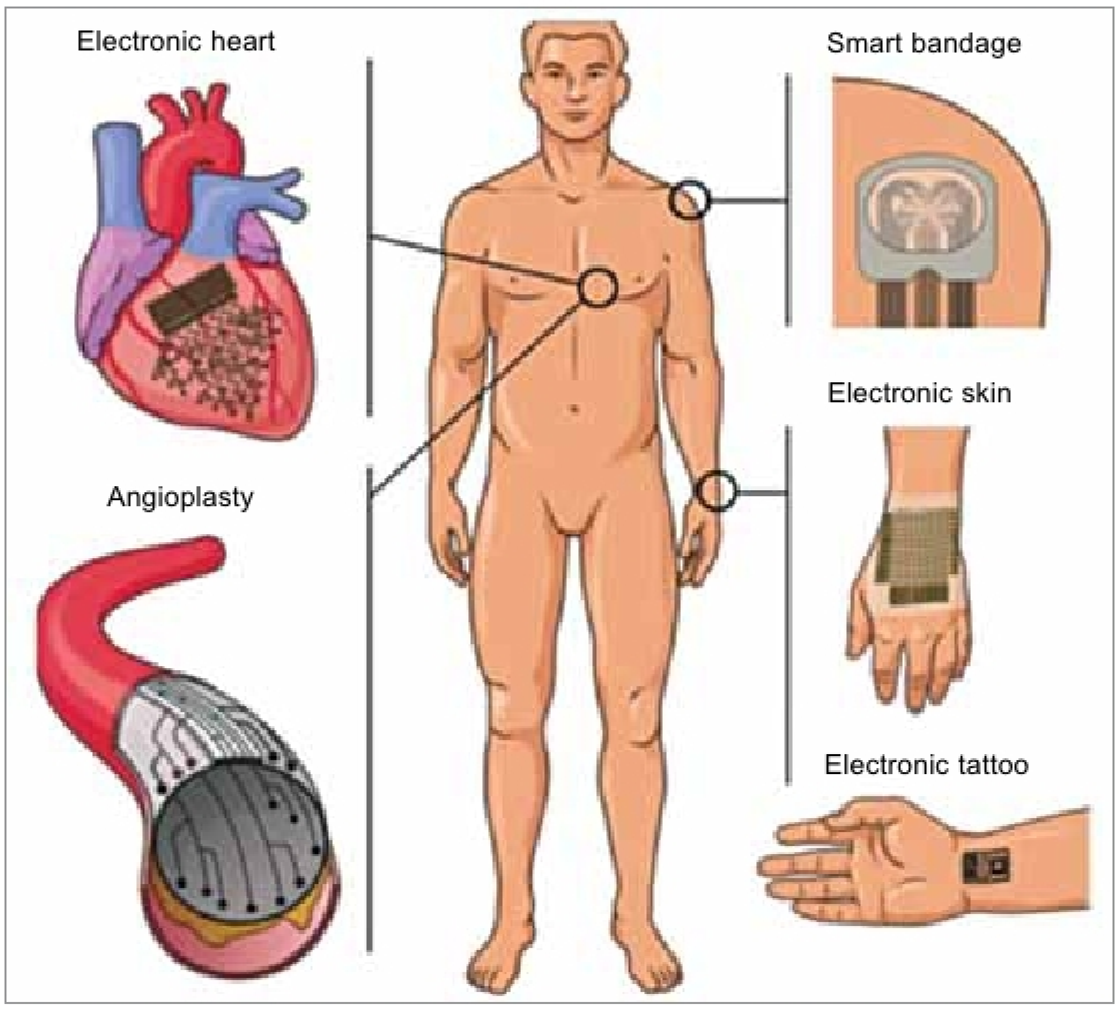

Fig. 1: Use of electronic devices

Further, such advances in electronic or artificial organs have been possible by the emergence of flexible and soft electronic materials that can readily integrate with the human body. An electronic organ is a device or tissue that is integrated into the human body—interfacing with living tissues—to replace a natural organ in order to duplicate or augment a specific function. This helps patients return to normal life as soon as possible. Use of artificial organ may help patients in the following ways:

- By providing life support to prevent imminent death while awaiting a treatment
- By improving the ability for self-care
- By improving the ability to interact socially
- By improving the quality of life

This article covers the importance of electronics in medical sciences with special emphasis on the development of electronic devices, which not only treat injuries and diseases but also help replace infected human organs. Modern electronic devices are being developed with the use of smart and multifunctional materials to satisfy the need for soft, curvilinear, and continuously evolving characteristics of the human body.

The resulting electronic devices with unique capabilities in biomedical research and clinical medicine can very well integrate onto or into the human body for diagnostics, therapeutic, or surgical functions with a strong potential to improve human health